Cypress Testing & Integration Guide

1. Introduction to Cypress

Cypress is a modern **end-to-end testing framework** used to test web applications. It allows developers and QA engineers to simulate real user behavior such as clicking buttons, filling forms, and navigating pages.

Key Features:

- Fast and reliable testing
- Runs directly in the browser
- Automatic waiting and retry mechanism
- Built-in debugging tools
- Supports CI/CD integration

2. Creating and Running a Cypress Project

Step 1: Create Project

```
mkdir cypress-project
cd cypress-project
npm init -y
```

Step 2: Install Cypress

```
npm install cypress --save-dev
```

Step 3: Open Cypress

```
npx cypress open
```

Select **E2E Testing** and choose a browser.

Step 4: Create a Test

cypress/e2e/sample.cy.js

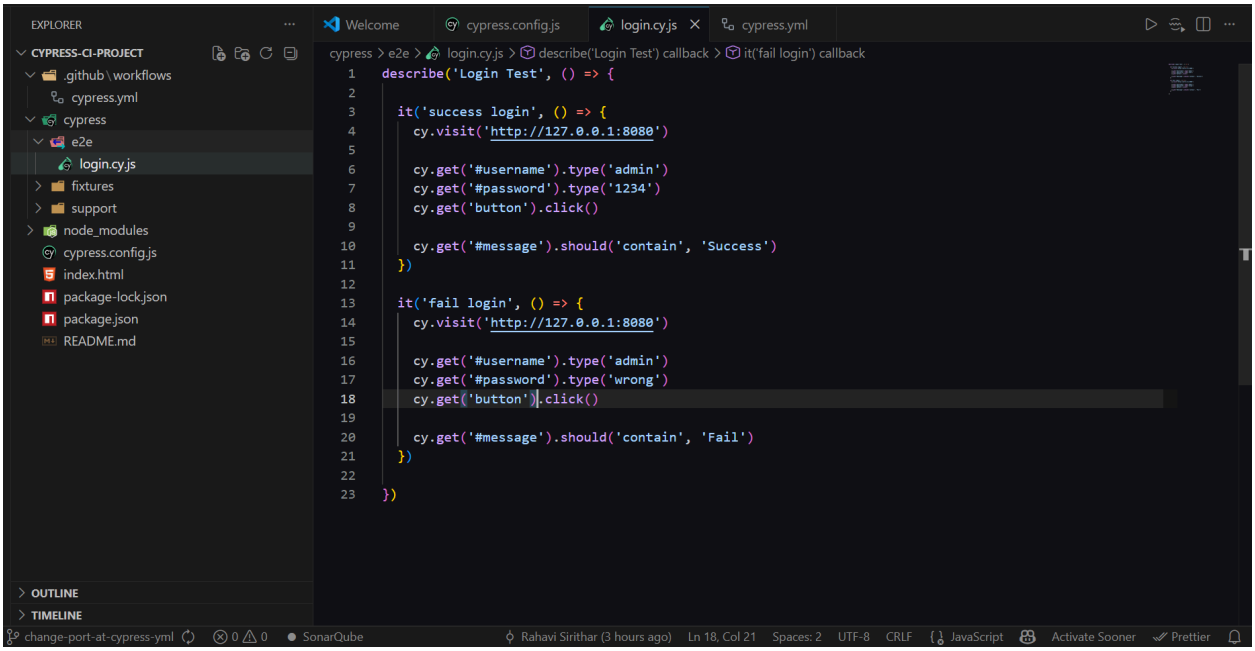
```
describe('Sample Test', () => {
```

```
  it('Visits a website', () => {
    cy.visit('https://example.cypress.io')
    cy.contains('type').click()
  })
})
```

Step 5: Run Tests

npx cypress run

Cypress provides two modes: interactive mode for development (npx cypress open) and headless mode for automation (npx cypress run).



3. Cypress Cloud Integration

Cypress Cloud provides:

- Test recordings (videos, screenshots)
- Dashboard for results
- Parallel test execution
- Debugging tools

Documentation:
https://docs.cypress.io/cloud/integrations/github

4. GitHub Integration (CI/CD)

Why integrate with GitHub?

- Automatically run tests on every push
- Prevent broken code from merging
- Display test results in pull requests

Step 1: Create GitHub Repository

Go to: GitHub

Step 2: Push Code

```
git init
git add .
git commit -m "Initial commit"
git remote add origin <repo-url>
git push -u origin main
```

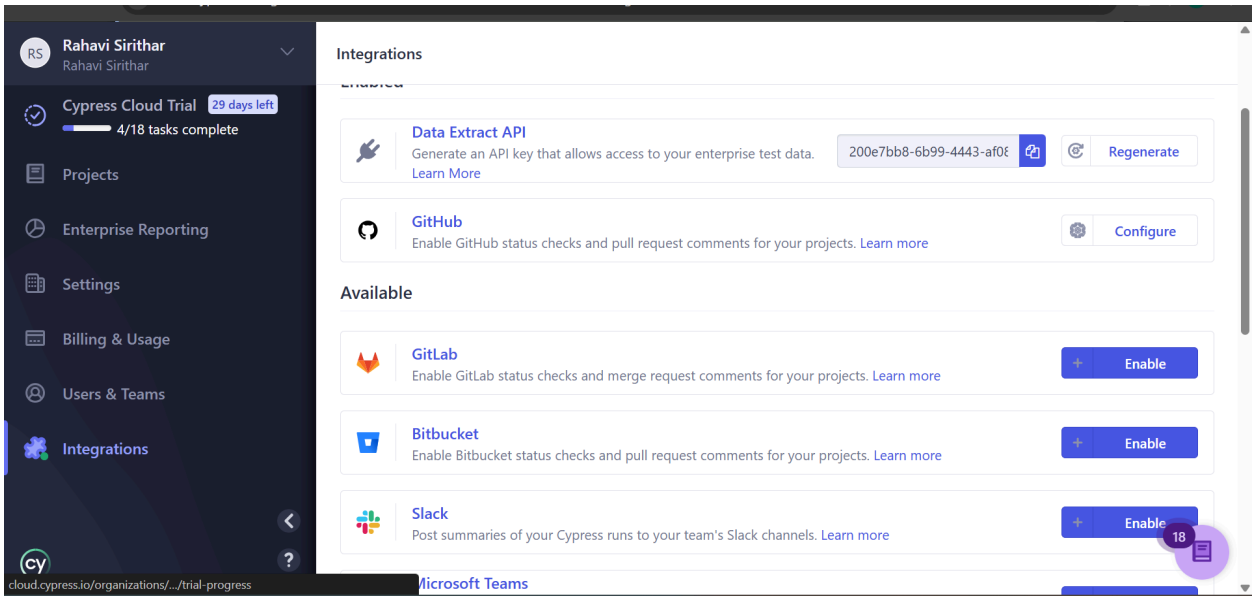
Step 3: Add GitHub Actions

.github/workflows/test.yml

```yaml
name: Cypress Tests

on: [push, pull_request]

jobs:
  test:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3

      - uses: actions/setup-node@v3
        with:
          node-version: 18

      - run: npm install

      - run: npx cypress run
```

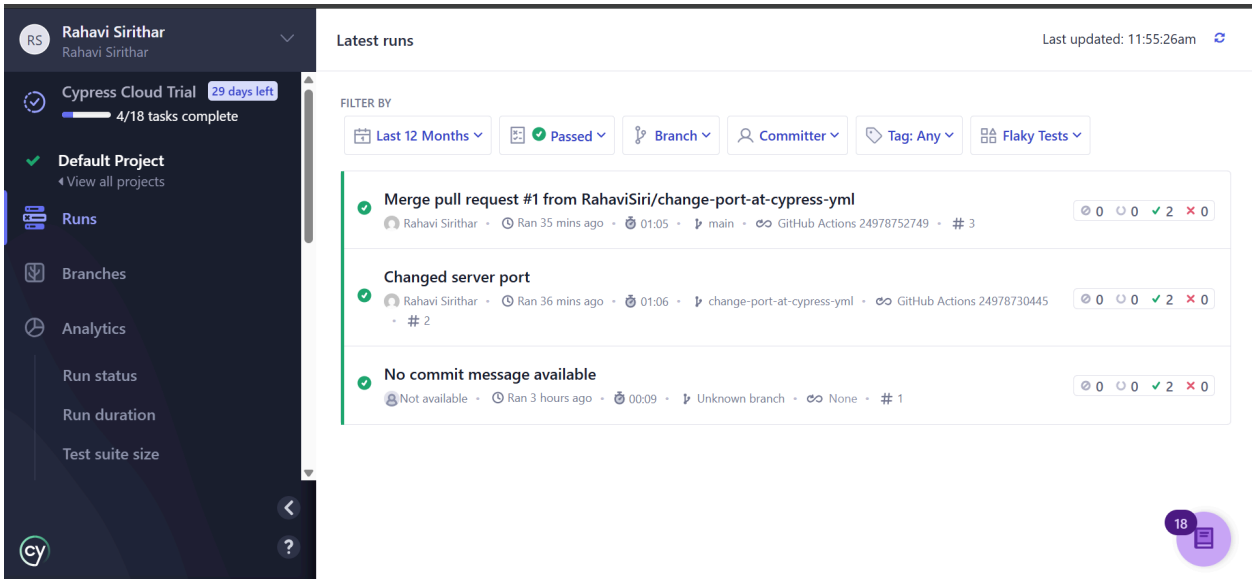
Step 4: Enable Cypress Cloud GitHub Integration

1. Go to Cypress Cloud
2. Open **Integrations**
3. Click **Enable GitHub**
4. Select your repository



Result:

- GitHub shows test status (pass/fail)
- Pull request comments
- Detailed logs in Cypress Cloud



5. Where Else Cypress Can Integrate

Cypress supports integration with multiple tools:

1. Git Platforms

- GitHub
- GitLab
- Bitbucket

Used for:

- Code management
- CI/CD pipelines

2. Communication Tools

- Slack
- Microsoft Teams

Used for:

- Sending test results notifications
- Alerting teams on failures

3. Bug Tracking Tools

- Jira

Used for:

- Automatically creating bug tickets when tests fail

4. CI/CD Tools

- GitHub Actions
- Jenkins
- CircleCI

Used for:

- Automating test execution

6. Why Integrations Are Important

Integrations help in:

- Automation → tests run automatically
- Error prevention → stop broken code
- Team collaboration → everyone sees results
- Faster development → quick feedback
- Debugging → logs, videos, screenshots

7. Conclusion

Cypress is a powerful testing tool that becomes even more effective when integrated with platforms like GitHub and CI/CD tools. It helps ensure code quality, reduces bugs, and improves development efficiency through automation and real-time feedback.

8. Summary Flow

Write test → Push code → CI runs Cypress → Results in Cypress Cloud → GitHub shows status